

Cost-Aware Conditional Diffusion Compression of 4K-5K Drone Imagery on GPU HPC Systems

Jooho Kim, Anish Shakya, Yifan Yang, and Jacob Kelly
Texas A&M University, College Station, TX, USA

Abstract—High-resolution drone imagery gives disaster-response teams street- and parcel-scale visual evidence, but it also creates a practical HPC bottleneck. Each cropped RGB image in our workflow has size 5440×3648 , so repeated compression, reconstruction, storage movement, and visual inspection become expensive before downstream analysis begins. This summary studies conditional diffusion compression (CDC) as an operating-point problem for GPU systems. We ask which runtime controls make an existing CDC workflow usable for large disaster-imagery collections.

I. Summary

The workflow compresses full-resolution drone images with a pretrained x-parameterized CDC model and reconstructs them with a diffusion decoder on NCSA DeltaAI. Each experiment records wall time, peak GPU memory, bits per pixel (BPP), compression ratio, PSNR, SSIM, high-percentile error, and tiling seam metrics. We report compression ratio against uncompressed 8-bit RGB storage:

$$R = \frac{24}{\text{BPP}}. \quad (1)$$

For a candidate setting θ , we track runtime and memory gains relative to a no-tiling baseline:

$$S_T(\theta) = 1 - \frac{T_\theta}{T_{\text{base}}}, \quad G_M(\theta) = \frac{M_{\text{base}}}{M_\theta}. \quad (2)$$

These equations make the systems trade-off explicit; they do not replace image-quality review.

The native full-image baseline is feasible but costly. With 65 denoising steps, fp32 reporting, and batch size 1, the DeltaAI GH200 run takes about 144 seconds per image and peaks near 52 GB of GPU memory. Step reduction speeds decoding but does not remove the memory footprint. Resolution reduction is faster, but it discards spatial detail before compression. In the 20-image sweep, the 2K setting lowers runtime to about 7.36 seconds per image, yet SSIM drops to about 0.689, which is too aggressive for detailed visual inspection without downstream validation.

Tiling gives the strongest current operating point because it keeps the native image boundary while reducing the tensor size seen by the decoder. On 50 full-resolution images, 256×256 tiling lowers runtime from 144.35 to 79.84 seconds per image and peak memory from 52.01 to 1.66 GB. This is a 44.7% runtime reduction and a $31.3\times$ peak-memory reduction. It is also faster and smaller than 512×512 tiling, with only small PSNR and SSIM losses.

TABLE I

Selected DeltaAI GH200 tiling validation, 50 full-resolution images, 65 denoising steps, fp32.

Setup	Sec./img	GPU GB	PSNR	SSIM
No tiling	144.35	52.01	28.96	0.8616
256^2 tile	79.84	1.66	28.78	0.8552
512^2 tile	87.71	3.02	28.86	0.8583

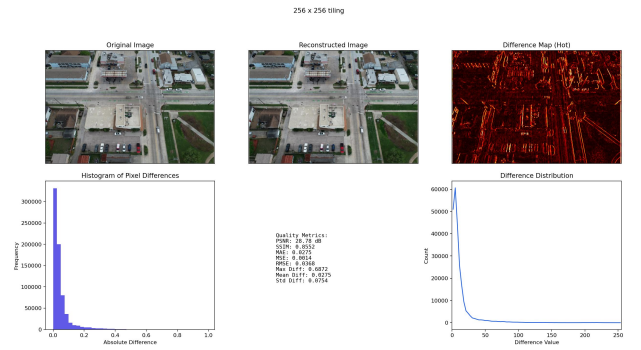


Fig. 1. Existing experiment output for the selected 256×256 tiling case: original preview, reconstruction, difference map, histogram, metrics, and difference distribution.

The main lesson is that CDC deployment for disaster imagery must be reported as a joint systems and image-quality problem. BPP and compression ratio are not enough. A useful setting must fit GPU memory, reconstruct quickly, preserve inspection details, and expose artifacts through comparisons and heatmaps. The current evidence supports 256×256 tiling as the best speed-memory candidate for the poster draft, with final-version updates planned for a larger sample, more visual panels, and downstream checks.

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